Terminal User Interfaces

What's really happening when you run htop

Nirmal Kumar

I was browsing reddit one night…

All this - On a terminal?

```
# Prefix
set -g prefix C-a

# General
set -g mouse on
set -g status-position bottom
set -g default-terminal "tmux-256color"
set -g set-titles-string "#S:#I.#P"
set -g allow-rename off
set -g automatic-rename off
set -g base-index 1
setw -g pane-base-index 1

# Key bindings
bind r source-file ~/.tmux.conf
bind-key x kill-pane
bind-key m split-window -v
bind-key u split-window -h
bind-key Tab last-window
unbind l

# Plugins
set -g @plugin 'tmux-plugins/tpm'
set -g @plugin 'tmux-plugins/tmux-sensible'
set -g @plugin 'tmux-plugins/tmux-cpu'
set -g @plugin 'tmux-plugins/tmux-battery'
set -g @plugin 'catppuccin/tmux#v2.1.3'
set -g @plugin 'christoomey/vim-tmux-navigator'
set -g @vim_navigator_prefix_mapping_clear_screen ""
bind C-l send-keys 'C-l'

# Catppuccin (must be set before TPM initializes plugins)
set -g @catppuccin_flavor 'mocha'
set -g @catppuccin_window_status_style "rounded"
set -g @catppuccin_window_flags "icon"
set -g @catppuccin_window_text " #W "
set -g @catppuccin_window_current_text " #W "

# Initialize TPM (must come before plugin run commands)
run '~/.tmux/plugins/tpm/tpm'

# Status bar
set -g status-justify left
set -g status-right-length 100
set -g status-left-length 100
set -g status-left ""
set -g status-right "#{E:@catppuccin_status_directory}"
set -ag status-right "#{E:@catppuccin_status_application}"
set -ag status-right "#{E:@catppuccin_status_session}"
set -agF status-right "#{E:@catppuccin_status_cpu}"
set -agF status-right "#{E:@catppuccin_status_battery}"
```

```
NORMAL  main  tmux/.tmux.conf   A  5   41%  23:33  08:08
1 Main   2 Lazygit
```

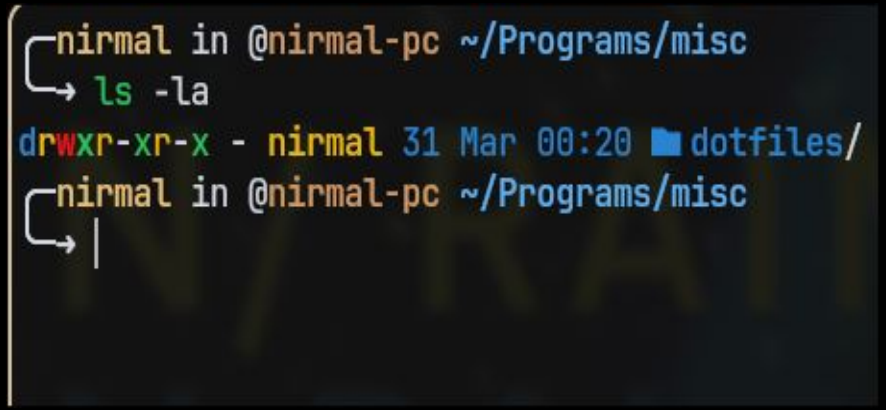
```
08:07
❯ neofetch
Macbook-Air-M1
: MacOS
: Apple M1
DE: Aqua
: 1440x900
: 58%
: yabai
: tmux
: spotify
```

```
cpu  menu  preset 2                    08:08:40   BAT▼ 58% 05:01  - 2000ms +
M1
CPU                    6%   41°C
C0   9%  C2   6%  C4  10%
C1   8%  C3   9%  C5   6%
Load avg: 3.86 3.07 3.05
up 12:52:10

mem                                    disks                          io
Total:                8.00 GiB         root                           228 GiB
Used                  3.30 GiB         IO%
Avail                 4.69 GiB         Used: 72%                      165 GiB
Cache                 1.83 GiB         VM                             228 GiB
Free                   920 MiB         IO%
                                       Used: 72%                      165 GiB
                                       Preboot                        228 GiB
                                       IO%

net  192.168.31.234                    sync  auto  zero  ←b en0 n→
10K                                    download
                                       ▼ 3.48 KiB/s (27.9 Kibps)
                                       ▲ 510 Byte/s (3.98 Kibps)
10K                                    upload
dotfiles  nvim  0  5.0%  58%
```

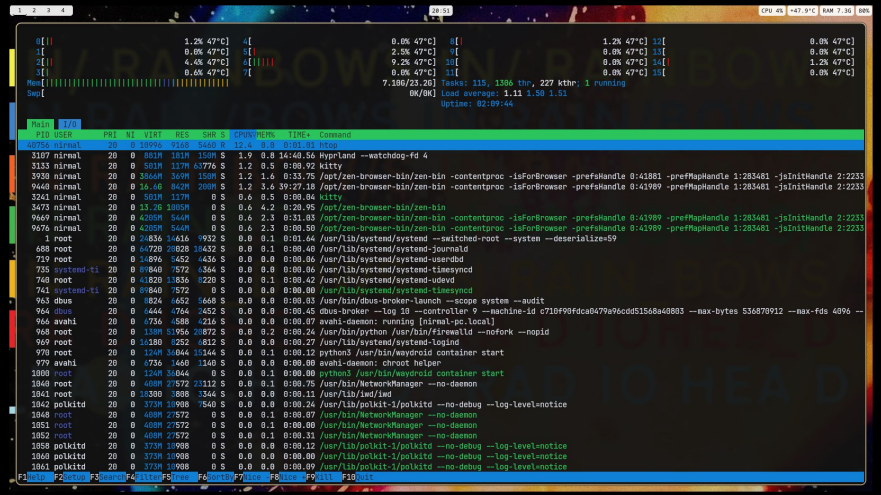
What is a TUI?

- Full fledged app living entirely in your terminal!
- Rendered completely by using characters and escape sequences!
- Eg : vim, htop, ranger, etc.,

CLI



TUI

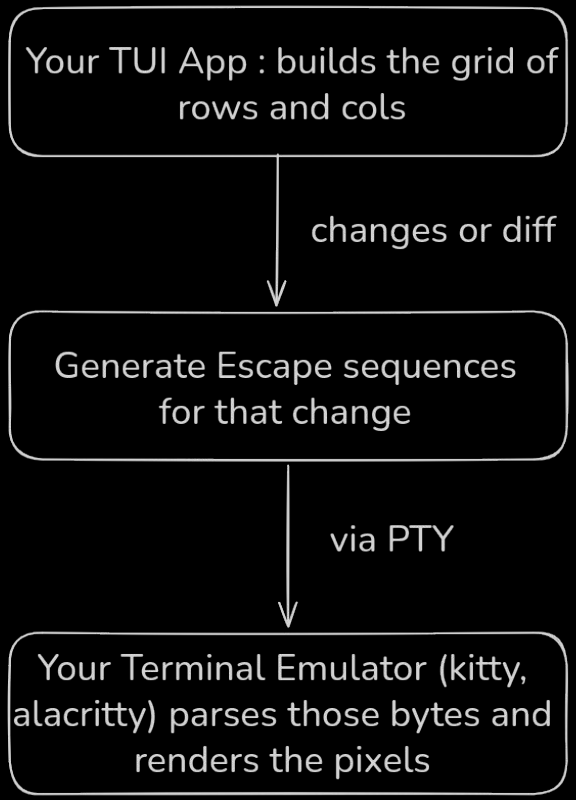


Terminal Rendering

- A terminal - 2D Buffer with rows and columns
- What does it send exactly? ANSI Codes!
- Runs a continuous loop.

ANSI Codes

"\033[2J"	clear screen
"\033[H"	move cursor to (0,0)



1 2 3 4 5 | 00:25 | CPU 3% | +50.6°C | RAM 10.8G | 79%

```
  0[|||||                    10.4% 50°C]   4[|||                    5.6% 50°C]   8[||||                   4.9% 50°C] 12[||                   3.1% 50°C]
  1[                          0.0% 50°C]   5[|||                    5.5% 50°C]   9[                      0.0% 50°C] 13[|                    0.6% 50°C]
  2[|||||                    10.5% 50°C]   6[||||||||              15.5% 50°C]  10[|||                   4.9% 50°C] 14[|||                  4.9% 50°C]
  3[||                        2.4% 50°C]   7[|                      0.6% 50°C]  11[                      0.0% 50°C] 15[                     0.0% 50°C]
Mem[||||||||
```

Few ANSI Examples

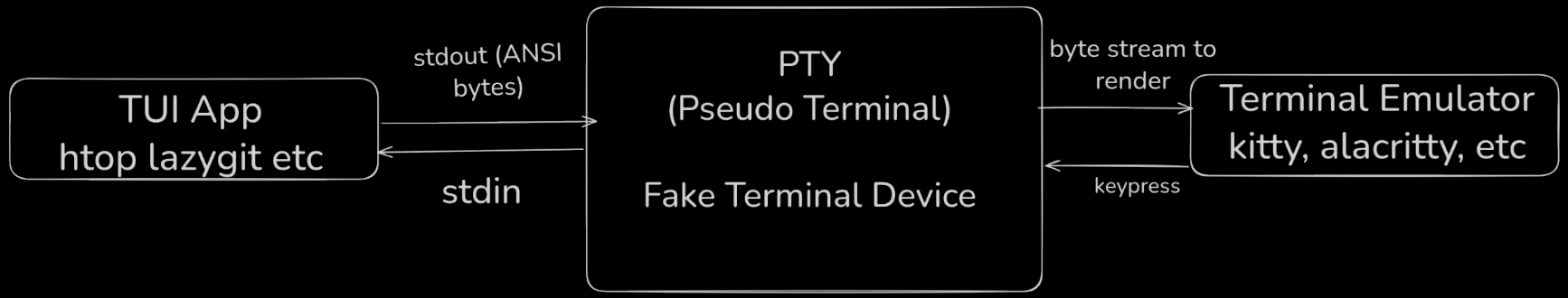
- Where :
  - ESC is \x1B

- So to move the cursor to home position :
  - \x1B[H

ESC Code	What it does
`\x1B[H`	move cursor to home (0,0)
`\x1B[2J`	clears the screen
`\x1B[31m`	set text color to red
`\x1B[1m`	bold text

```
┌nirmal in @nirmal-pc ~
└→ echo "\x1B[31mHello\x1B[0m"
Hello
```

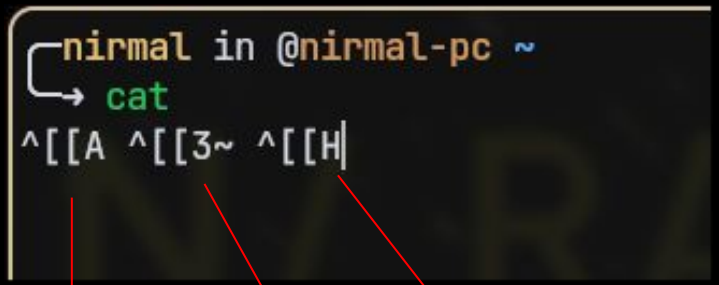
The PTY



- The Pseudo Terminal (PTY) : software layer that connects your terminal emulator to your TUI program.
- Fake Terminal Device
- Input and output byte streams are handled by the PTY

Raw Mode v/s Cooked Mode

- We need continuous input with no buffering.
- So we take input byte by byte.



Up Arrow

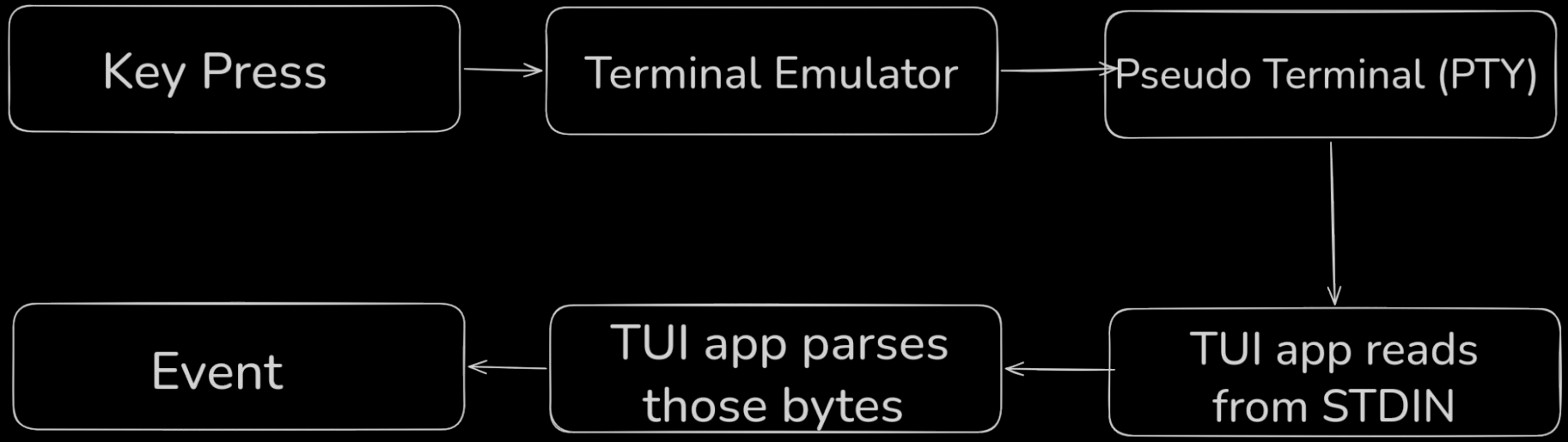
Delete Key

Home Key

Cooked Mode	Raw Mode
Default used by Terminal Emulators	Used in TUIs
Buffered Input	Input is taken as a byte stream
No Control	The program decides what to do with the bytes

Input Handling

Key press becomes structured events via byte streams



Practical Use Cases

- Archinstall
- btop/bashtop
- Low end hardware support

```
> desktop: Provides a selection of desktop environments and tiling window managers, e.g. gnome, kde, sway
  minimal: A very basic installation that allows you to customize Arch Linux as you see fit.
  server: Provides a selection of various server packages to install and enable, e.g. httpd, nginx, mariadb
  xorg: Installs a minimal system as well as xorg and graphics drivers.
(Press "/" to search)
```

```
Select your desired desktop environment

Use ESC to skip
  awesome
  bspwm
  budgie
  cinnamon
  deepin
  enlightenment
  gnome
  i3
  kde
  lxqt
  mate
  qtile
  sway
> xfce4
(Press "/" to search)
```

```
Select a graphics driver or leave blank to install all open-source drivers

Use ESC to skip
  AMD / ATI (open-source)
> All open-source (default)
  Intel (open-source)
  Nvidia (open kernel module for newer GPUs, Turing+)
  Nvidia (open-source nouveau driver)
  Nvidia (proprietary)
  VMware / VirtualBox (open-source)
(Press "/" to search)
```

Trade Offs

- Beautiful GUIs often dependent on the terminal emulator themselves.

- Each pixel in a character, so smooth non aliased text lines or elements is difficult to produce, unlike GUIs.

- Input Constraints

Thank You :D